nature machine intelligence

Article

https://doi.org/10.1038/s42256-026-01309-6

Towards foundation-style models for energy-frontier heterogeneous neutrino detectors via self-supervised pretraining

Received: 9 April 2026

Accepted: 26 August 2026

Published online: 30 September 2026

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Accelerator-based neutrino physics is entering an energy-frontier regime in which interactions reach the tera-electronvolt scale and produce exceptionally dense, overlapping detector signatures. Event interpretation is impractical for conventional reconstruction and challenging for supervised models trained from scratch, particularly when labelled data are scarce and analyses span diverse objectives. Here we present a sparse Vision Transformer framework for learning reusable representations from heterogeneous detector data. Self-supervised pretraining combines masked-autoencoder reconstruction with relational voxel-level objectives for hierarchy, ghost and particle identification, followed by joint fine-tuning across classification and regression tasks. On simulated events from the proposed FASER calorimeter (FASERCAL) concept, pretraining improves neutrino-flavour and charm-quark identification, momentum regression and vertex reconstruction, with relational objectives providing further gains in topologically complex channels. Attribution and representation diagnostics show a more structured latent space, while detector-subsystem ablations recover physically plausible channel-dependent roles. With roughly 10^3 labelled events, the pretrained encoder matches the flavour-classification performance of scratch training with an order of magnitude more data. Cross-domain transfer is observed on public plastic-scintillator and liquid-argon benchmarks. A matched alternative-generator stress test shows stable flavour and kinematic performance while exposing charm tagging as generator sensitive. These results support self-supervised multimodal pretraining as a route towards reusable detector representations. We use 'foundation-style' in this restricted sense, rather than claiming a completed general-purpose detector foundation model.

Accelerator-based neutrino physics is entering the energy frontier. Forward collider-neutrino programmes extend measurements into the tera-electronvolt regime, enabling neutrino-interaction studies and cross-section measurements at previously unexplored energies while broadening the case for collider neutrinos as a high-energy physics programme[1–6]. Yet, the same regime produces detector data of exceptional complexity. Interaction topologies become highly collimated, particle multiplicities rise, electromagnetic and hadronic activity overlap strongly and information must be integrated across heterogeneous detector systems. Machine learning is already widely used across particle physics for detector inference, anomaly detection or event reconstruction, and that broader landscape has been reviewed extensively[7–12]. In energy-frontier neutrino detection, however, the challenge is not simply whether learned models outperform existing pipelines but whether any practical analysis of these events is feasible without them.

Within accelerator neutrino experiments, convolutional, graph-based and transformer-like models have been shown to improve

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event classification, semantic segmentation or clustering on simulated detector data[13–21]. Most of literature, however, concerns lower-energy environments, individual detector subsystems or task-specific supervised models trained from scratch, often in settings where conventional reconstruction remains plausible. The regime considered here is qualitatively different: event topologies are so dense, collimated and overlapping that the challenge is not to refine an existing analysis chain but to construct a viable one.

The FASER calorimeter (FASERCAL) concept, a proposed upgrade for the FASER experiment at CERN, provides a stringent case study for this problem[22]. Its highly granular three-dimensional (3D) calorimeter (3DCAL) comprises over 460,000 readout voxels, only a fraction of which are active in a given event, and is followed by electromagnetic (ECAL) and hadronic (AHCAL) calorimeters and a muon spectrometer. The forward neutrino beam contains ν_e, ν_μ and ν_τ components, and the analyses considered here target both charged-current (CC) and neutral-current (NC) interactions. A model must therefore integrate sparse 3D volumetric inputs with heterogeneous auxiliary streams of different dimensionality. The resulting events, shown in Fig. 1, contain dense shower cores, extended secondary tracks, partial containment and deeply ambiguous local configurations. In this setting, machine learning-based analysis is not an optional enhancement to an otherwise tractable reconstruction problem; it is a mandatory prerequisite for practical event-level physics extraction.

This is precisely the setting in which representation learning becomes compelling. High-fidelity simulation provides abundant detector responses, but labelled targets remain task specific and expensive to produce. Masked modelling has emerged as a strong route to transferable representations in language, vision and structured data[23–28]. In neutrino and particle physics, recent studies have begun exploring transfer learning, domain adaptation, contrastive learning and masked point modelling[29–36]. What remains missing is a unified approach for learning reusable models for genuinely heterogeneous detectors in the extreme topological regime of energy-frontier neutrino interactions.

Our aim is to take a concrete step towards foundation-style models for neutrino detector data. We do not claim a completed general-purpose model; instead, we target the core ingredients such a model requires: broad self-supervised pretraining; reused across several downstream tasks; strong performance when only a few hundred to a few thousand labelled events are available; and transfer across detector technologies and energy scales. In this manuscript, 'foundation-style' therefore has an operational meaning: a single reusable encoder is trained with objectives independent of downstream task labels on a large detector sample, adapted to several supervised physics tasks, tested under limited-label conditions, probed with a matched alternative-generator sample and reused outside the source detector. A full detector foundation model would require broader detector coverage, stronger systematic validation and ultimately experimental-data calibration. We address this with a sparse vision-transformer (ViT)-like[37] framework that combines masked-autoencoder (MAE)[24] pretraining with relational voxel-level objectives available from simulation. The encoder is later fine-tuned on flavour identification, charmed-quark identification, event kinematics and vertex reconstruction; stress-tested on matched GENIE and NuWro samples; and evaluated for transfer on public datasets outside the source domain.

This paper makes three main contributions.

(1) We introduce a sparse encoder for heterogeneous detector data that combines sparse convolutional patch embeddings, module-aware self-attention and Perceiver-IO fusion across calorimetric and tracking streams.
(2) We formulate a multimodal pretraining strategy that augments masked reconstruction with relational voxel-level targets (ghost identification, interaction hierarchy and particle category), and show in the in-domain comparisons that include all three baselines that this composite objective improves downstream performance beyond MAE-only pretraining, with the largest gains in the most challenging channels.
(3) We demonstrate that the learned representation improves performance and data efficiency across a multi-task fine-tuning suite and transfers beyond the source domain to publicly available benchmarks spanning different detector technologies and energy regimes. We also include a matched GENIE-to-NuWro evaluation of the final MAE + Rel to probe generator dependence without retraining and report the cross-domain transfer summaries for all three initialization strategies.

We evaluate this framework first on simulated events from the FASERCAL concept as an energy-frontier case study, including a matched GENIE-to-NuWro stress test, and then on public transfer benchmarks covering a variety of detector technologies and energy scales. The 'Results' therefore moves from pretraining behaviour to downstream classification, regression, attribution and representation diagnostics; data efficiency; and a combined robustness-and-transfer evaluation. The 'Discussion' interprets what these findings imply for representation learning in detector physics and the Methods describe the case-study data, architecture, training strategy and transfer setup.

Results

All downstream comparisons use the same architecture under three initialization strategies:

1. Scratch: the fine-tuning architecture with standard random initialization of all trainable parameters.
2. MAE: the same downstream architecture and randomly initialized task heads, with the encoder initialized from masked-reconstruction pretraining.
3. MAE + Rel: the same downstream architecture and randomly initialized task heads, with the encoder initialized from the full pretraining objective that combines masked reconstruction with voxel-level relational tasks.

Throughout the downstream results, Scratch, MAE and MAE + Rel denote the fully fine-tuned multi-task models, not frozen encoders.

Pretraining

The pretraining tasks are intended to shape the latent representation rather than to serve as end products, so we present them primarily as qualitative evidence. Figure 1 shows masked-reconstruction examples on test events. Even with 75% of occupied patches masked, the model recovers broad shower envelopes, elongated track-like structures and coherent energy flow into missing regions. The reconstructions are not exact, particularly in dense shower cores and around fine secondary structures, and we do not interpret them as precise generative predictions. Their value is diagnostic: the encoder is forced to infer non-local spatial correlations rather than copying visible voxels. The relational objectives are also shown in Fig. 1. The model predicts ghost labels, hierarchy labels and particle-category labels on kept patches, again on test events. Here, ghost voxels are reconstructed deposits with no matched true particle, hierarchy labels distinguish background, primary and secondary activity, and particle-category labels separate electromagnetic, muonic and hadronic deposits. These targets are especially demanding: a single reconstructed voxel can receive contributions from multiple true particles, yielding soft class mixtures rather than hard one-hot assignments. Agreement with ground truth is the strongest in well-separated tracks and along the dominant shower axes, while the densest overlapping regions remain the most difficult. This is consistent with the detector challenge described in ref. 22: dense occupancy and view ambiguities create ghost activity and complicate local pattern assignment. Taken together, the reconstruction and relational results in Fig. 1 suggest that the encoder captures meaningful spatial and semantic correlations before any downstream fine-tuning is applied.

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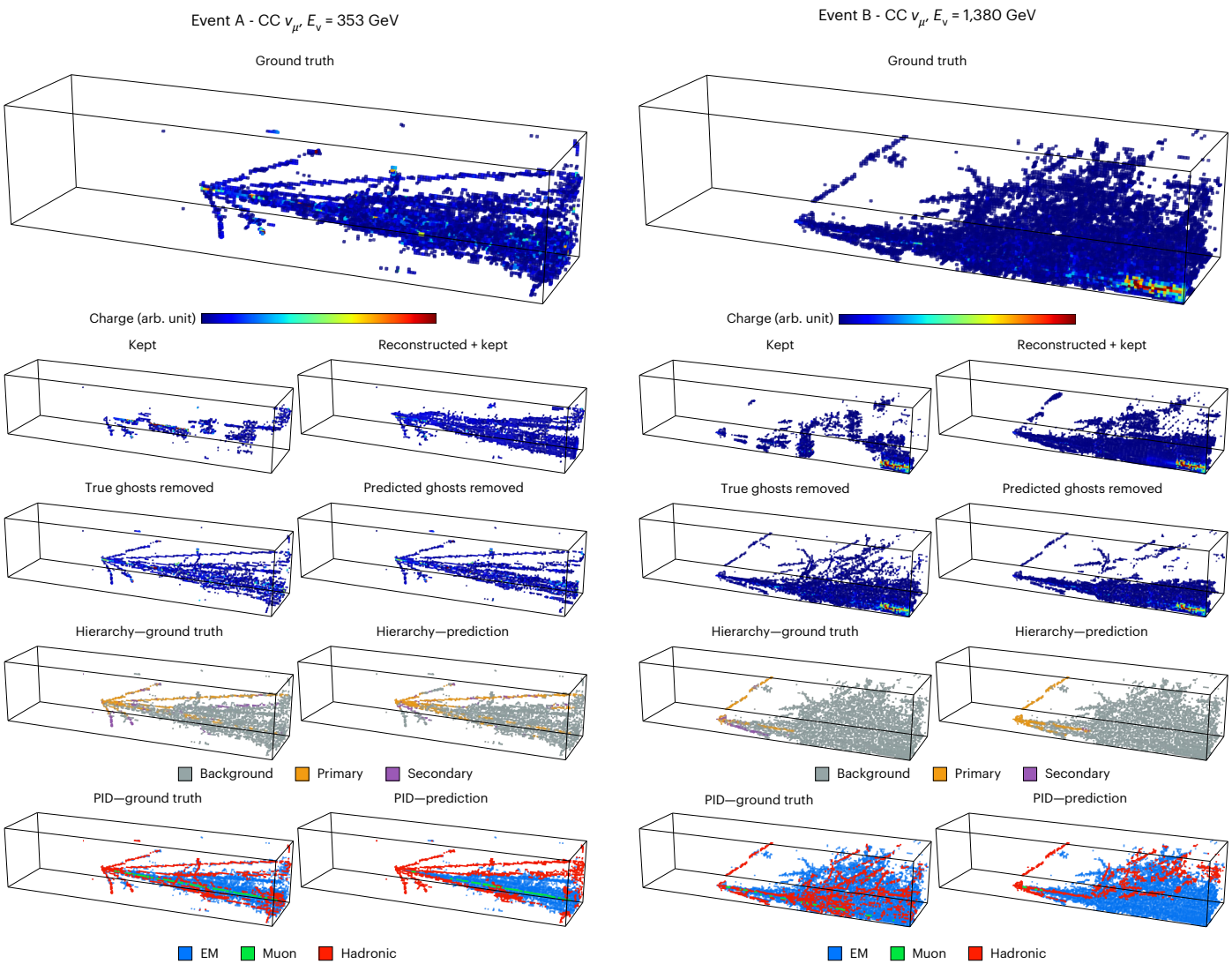

Fig. 1 | Example events from the test set illustrating masked reconstruction and relational tasks in the 3DCAL detector. For each event, the top row displays the ground-truth detector readout. The second row presents the visible voxels in the masked input (25% kept) and the reconstructed output combined with the visible voxels (reconstructed + kept). The third row compares ghost-hit removal using ground truth and model predictions. The fourth row illustrates interaction hierarchy labelling (background, primary and secondary). The bottom row presents particle identification (electromagnetic, muon and hadronic). For the hierarchy and particle-identification panels, each voxel is assigned to its dominant contributing class for visualization.

Classification and regression

The downstream classification and regression results are shown together in Fig. 2. The figure combines the most diagnostic classification panels with an all-class regression summary, while the operating-point scans and class-resolved regression box plots are shown in Supplementary Figs. 1 and 2. Performance is evaluated with one-versus-rest receiver operating characteristic (ROC) curves, row-normalized confusion matrices and the figure of merit FOM $= S/\sqrt{S+B}$, where S and B denote the expected signal and background yields after thresholding. For flavour classification, CC ν_τ events are split into $\tau \to e$, $\tau \to \mu$ and $\tau \to$ had according to whether an electron or muon appears among the primary tau-decay products; all remaining CC ν_τ events are assigned to the hadronic class. For charm classification, events are assigned to charm $\to \mu$ if any primary charm-decay product is a muon, to charm $\to e$ otherwise if any primary charm-decay product is an electron and to charm $\to$ had for the remaining charm decays. The overall pattern is clear. MAE already improves the dominant channels, while MAE + Rel provides the largest gains for the less abundant and more topologically complex signatures.

For the dominant channels, the gains are consistent. The ν_e CC area under the ROC curve increases from 0.968 for Scratch to 0.983 for MAE and 0.985 for MAE + Rel, the ν_μ CC area under the curve (AUC) increases from 0.909 to 0.955 and 0.958, and the NC AUC increases from 0.885 to 0.943 and 0.947. The row-normalized confusion matrices also become more diagonal for the common categories. For example, the ν_e CC diagonal entry rises from 0.71 to 0.84 and 0.85, while the NC diagonal rises from 0.50 to 0.71 and 0.70. This suggests that masked reconstruction effectively captures the global event morphology needed for the dominant classes.

The most consequential gains appear in the tau-neutrino channels. The threshold scans supporting the quoted maximum figure of merit values are shown in Supplementary Fig. 1. For ν_τ CC $\to$ had, the AUC rises from 0.902 to 0.941 and 0.944, and the maximum figure of merit increases from 1.58 to 4.43 and 4.58. For ν_τ CC $\to e$, the AUC rises from 0.892 to 0.919 and 0.921, while the maximum figure of merit increases from 0.38 to 1.03 and 1.35. The muonic tau channel shows a similar trend, with the AUC improving from 0.801 to 0.831 and 0.835 and the maximum figure of merit from 1.25 to 1.79 and 1.82.

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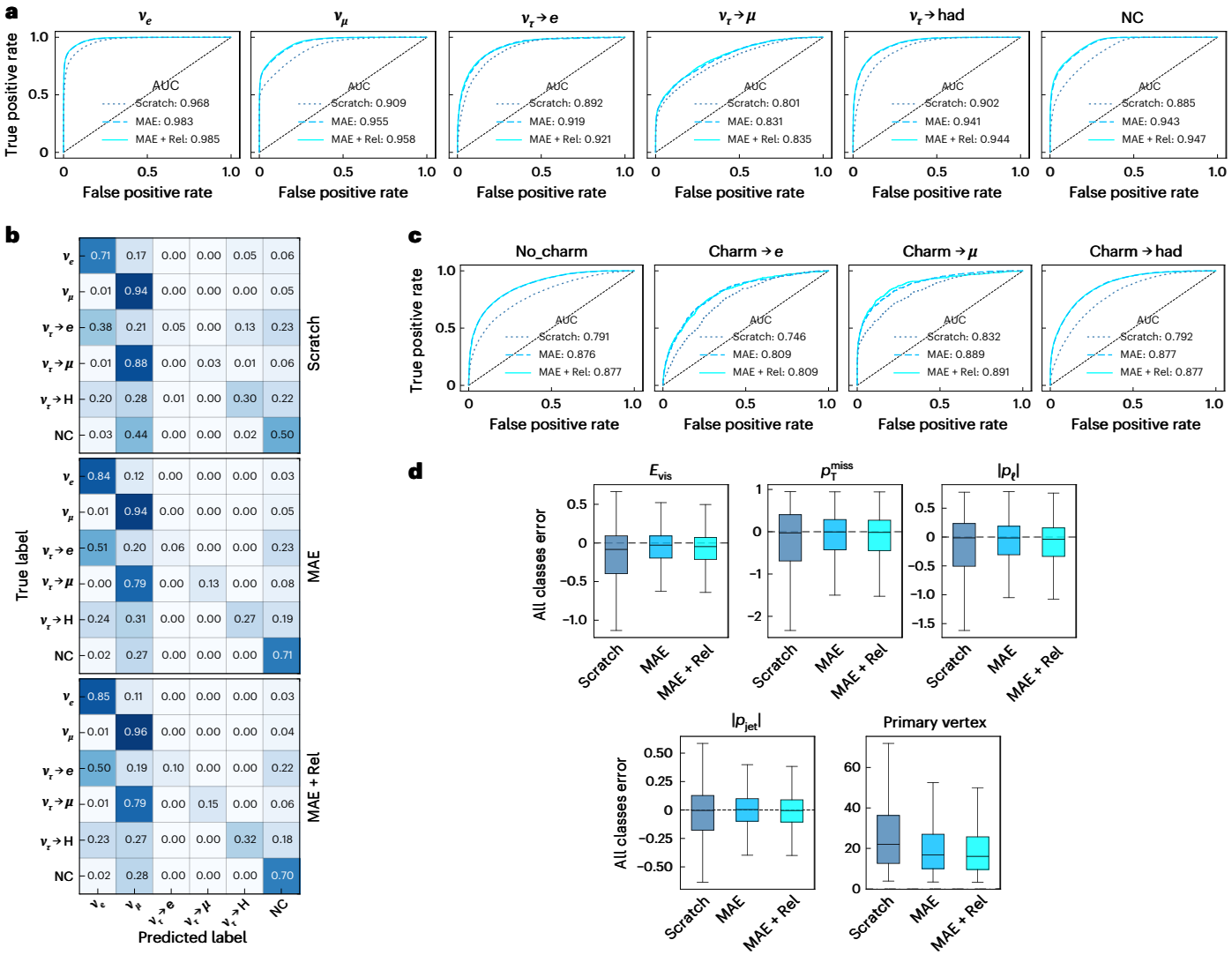

Fig. 2 | Classification and regression performance. a, One-versus-rest ROC curves for the six flavour categories, comparing Scratch, MAE and MAE + Rel. **b**, Row-normalized flavour confusion matrices using per-class thresholds that maximize the figure of merit; events passing multiple thresholds are assigned to the highest score. **c**, One-versus-rest ROC curves for charm categories. **d**, All-class regression-error summary, shown for E_{vis}, p_T^{miss}, $|p_\ell|$, $|p_{jet}|$ and the primary-vertex error. All panels use the 10% GENIE test partition ($n = 122{,}638$ independent simulated neutrino-interaction events); one event is the unit of study. Each event is evaluated once by each model. The groups compared are Scratch, MAE and MAE + Rel; Scratch is the randomly initialized baseline trained without self-supervized pretraining. In **d**, momentum and energy errors are fractional residuals $(x^{true} - x^{reco})/x^{true}$; the primary-vertex error is $d_{PV} = \| \mathbf{x}_{PV}^{true} - \mathbf{x}_{PV}^{reco} \|$ in millimetre. Box plots show event-level distributions: horizontal bars mark the median, box bounds mark the 25th and 75th percentiles and whiskers extend from the 2.5th to the 97.5th percentiles. The corresponding purity, efficiency and figure-of-merit scans, together with class-resolved regression box plots, are shown in Supplementary Figs. 1 and 2. had: hadronic; FOM: figure of merit $(S/\sqrt{S+B})$; S and B denote expected signal and background yields.

The confusion matrices show that these channels remain the most difficult, with substantial leakage into ν_e CC, ν_μ CC and NC categories, but the pretrained models separate them more cleanly than Scratch, especially once the relational objective is included. These are precisely the channels in which dense overlap, secondary activity and partial containment complicate classification, so the quantitative pattern is important: the benefit of pretraining is the most pronounced where event interpretation is the hardest.

Charmed-quark classification shows the same structure. The no-charm category is already separated well by all models, but the less abundant charm decays benefit substantially from pretraining. For charm → μ, the AUC increases from 0.832 to 0.889 and 0.891, and the maximum figure of merit rises from 6.97 to 7.61 and 7.90. For charm → had, the AUC increases from 0.792 to 0.877 for both pretrained variants, while the maximum figure of merit rises from 27.10 to 37.27 and 37.74. Even charm → e, which remains challenging, shows an AUC increase from 0.746 to 0.809 and a maximum-figure-of-merit increase from 1.75 to 2.08 and 2.20. Overall, the relational objectives do not simply nudge the model upwards. They disproportionately improve the lower-yield channels that are the most critical for the detector's physics reach.

The same trend extends beyond classification. Figure 2d shows the regression errors for the event visible energy E_{vis}, the missing transverse momentum p_T^{miss}, the magnitudes of the primary-lepton and hadronic-jet momenta $|p_\ell|$ and $|p_{jet}|$, and the primary-vertex position after pooling the selected events across flavour categories. Primary-vertex performance is summarized with the 3D error $d_{PV} = \| \mathbf{x}_{PV}^{true} - \mathbf{x}_{PV}^{reco} \|$. Performance is best assessed through shifts in the medians and reductions in spread across the box plots, with class-resolved regression box plots shown in Supplementary Fig. 2.

The clearest and most uniform effect appears in primary-vertex reconstruction. In the all-class summary, the d_{PV} distributions for the

pretrained models are shifted towards smaller errors than Scratch and their interquartile ranges are reduced. The class-resolved box plots in Supplementary Fig. 2 show that this behaviour is not driven by a single easy topology but appears across the common ν_e CC and ν_μ CC channels as well as the more difficult tau and NC samples. Between the two pretrained variants, MAE + Rel is typically the most compact or among the most compact distributions.

The kinematic targets show a more heterogeneous but still favourable pattern. For E_{vis} and $|p_{\mathrm{jet}}|$, the pretrained models generally move the medians closer to zero and reduce the spread, with the clearest gains in the CC channels and in ν_τ CC → had, where jet reconstruction is especially challenging. For $p_{\mathrm{T}}^{\mathrm{miss}}$, the improvement is the most evident in the dominant channels, while the tau categories remain broad, reflecting the intrinsic difficulty of the selected sample. The primary-lepton momentum error also tends to tighten under pretraining where that observable is defined, although the gain is less uniform than for d_{PV}. Taken together with the classification results, Fig. 2 confirms that pretraining does not merely sharpen the final classifier head: it improves the shared latent representation underlying both discrete and continuous physics targets.

Attribution and representation diagnostics

Figure 3 provides complementary evidence about what the fine-tuned encoder is using and how pretraining shapes the latent space. We treat these analyses as attribution and representation diagnostics, not as a complete physical interpretation of the learned model. They identify detector regions, detector branches and latent-space organization associated with the predictions, while causal physics interpretation requires additional controlled studies. In Fig. 3a, patch-level 3DCal saliency maps for two arbitrary test CC events show that attribution is concentrated near the interaction region and along a limited subset of downstream shower and track structures rather than being spread uniformly across all active patches. Even in the more crowded tera-electronvolt-scale event, the highest-saliency regions follow the main event skeleton, suggesting that the model integrates localized interaction cues with extended downstream topology rather than relying on diffuse correlations alone. This pattern is consistent with the physics content of the tasks: flavour, charm and vertex inference depend both on the primary interaction region and on downstream track/shower development, especially for muonic and hadronic topologies.

Figure 3b compares two-dimensional uniform manifold approximation and projection (UMAP) maps of the pooled latent representations for Scratch, MAE and MAE + Rel, coloured by true flavour and visible energy. The Scratch model already exhibits coarse organization, while the pretrained variants form cleaner low-dimensional structures, with better separated flavour groupings and smoother visible-energy progressions across the embedding. The coexistence of flavour clustering and energy ordering suggests that the representation organizes events by multiple correlated physical attributes rather than along a single task-specific axis. Residual overlap, especially between NC and ν_μ CC events and among the tau channels, is consistent with the confusions shown in Fig. 2, indicating that pretraining sharpens but does not erase the hardest physical ambiguities. As a qualitative projection, UMAP is not itself a performance metric but it is consistent with pretraining producing a more structured shared representation.

In Fig. 3c, we test how that representation uses the heterogeneous detector inputs. The result is clear: 3DCAL provides the backbone of the event interpretation. Removing it lowers flavour macro-area under the ROC curve (AUROC) from 0.928 to 0.806 and macro-AUROC for charmed-quark classification from 0.866 to 0.647, while the mean vertex error rises from 21 mm to 966 mm. The auxiliary branches nevertheless provide targeted complementary information. Dropping AHCAL most strongly affects hadronic and NC discrimination, reducing the one-versus-rest AUROC from 0.939 to 0.857 for ν_τ CC → had and from 0.939 to 0.854 for NC events. Dropping the muon spectrometer has the clearest effect on muonic channels, reducing the one-versus-rest AUROC from 0.949 to 0.916 for ν_μ CC and from 0.848 to 0.751 for ν_τ CC → μ and worsening the primary-lepton-momentum resolution. Removing ECAL degrades energy-related observables, with the visible-energy σ_{MAD} increasing from 0.207 to 0.417. These ablations should be read as showing which detector inputs the trained model uses for interactions whose vertices lie in 3DCAL, not as an intrinsic ranking of subsystem importance. As shower containment and leakage couple the downstream detector response, removing one detector branch from the network does not erase that subsystem's physical influence on the topology observed by the others. The ablation pattern therefore matches the detector design: 3DCAL dominates, but the downstream subsystems contribute information in physically plausible, channel-dependent ways. This is detector attribution rather than full physics interpretability: it identifies model sensitivity to each input branch, but it does not prove that the network has learned a human-readable reconstruction algorithm.

In Fig. 3d, we address a complementary question: whether the learned representation is fragile to coherent detector mismodelling. We apply a common global energy-scale shift to the calorimetric deposits in 3DCAL, AHCAL and ECAL at inference time and re-evaluate MAE + Rel, with paired bootstrap bands obtained by resampling the evaluated events across all scale points. In the current 50-batch scan, corresponding to 3,200 events, flavour macro-AUROC remains essentially unchanged at 0.9326–0.9331, charm macro-AUROC varies from 0.8926 to 0.8894 and d_{PV} changes by only about 0.13 mm across the full ±10% range. For the regression observables, we report the absolute change in the median signed fractional residual relative to the nominal scale point. This isolates the drift induced by the nuisance itself, rather than conflating it with any pre-existing calibration offset of the nominal model. Over the full scan, the largest observed drifts are 0.0092 for E_{vis}, 0.0065 for p_T^{miss}, 0.0047 for $|p_\ell|$ and 0.0097 for $|p_{\mathrm{jet}}|$. The scan is not a full detector-systematics programme, but it does suggest that the learned representation is not acutely brittle to a $\mathcal{O}(10\%)$ coherent calorimeter-scale bias. It should not be interpreted as covering generator, flux, channel efficiency or subsystem-specific detector uncertainties; those require dedicated nuisance studies before deployment in an experimental analysis.

Fig. 3 | Attribution and representation diagnostics for the learned representation. a, Two example patch-level 3DCal saliency maps for test CC events from the fine-tuned MAE + Rel. **b**, UMAP of the pooled latent representation for Scratch, MAE and MAE + Rel, coloured by true flavour (top) and visible energy, E_{vis} (bottom). **c**, Detector-subsystem ablation for MAE + Rel, evaluated on the full test set after suppressing each detector branch at token fusion. Metrics are one-versus-rest flavour AUROC, charmed-quark macro-AUROC, mean 3D vertex error, and σ_{MAD} for visible energy, missing transverse momentum, primary-lepton momentum and jet momentum. **d**, Response of MAE + Rel to coherent global calorimeter energy-scale shifts applied at inference time in 3DCAL, AHCAL and ECAL. The central curves show the value of each metric computed from the evaluated event sample at each scale point: flavour macro-AUROC, charm macro-AUROC, d_{PV} and the absolute change in the median signed fractional residual, $|\Delta\mathrm{median}[(x^{\mathrm{true}}-x^{\mathrm{reco}})/x^{\mathrm{true}}]|$, for E_{vis}, $p_{\mathrm{T}}^{\mathrm{miss}}$, $|p_\ell|$ and $|p_{\mathrm{jet}}|$, always measured relative to the nominal scale point. In **d**, $n = 3$, 200 independent simulated neutrino-interaction events are used at each energy-scale point, corresponding to 50 batches of 64 events; one event is the unit of study. The groups compared are the inference-time calorimeter energy-scale values, with the nominal scale as the control. Shaded bands show paired bootstrap percentile intervals obtained by resampling the same events jointly across all scale points; bootstrap resamples are a statistical uncertainty estimate. The panels should be read as diagnostic probes of model reliance and latent organization rather than exhaustive explanations of the learned physics.

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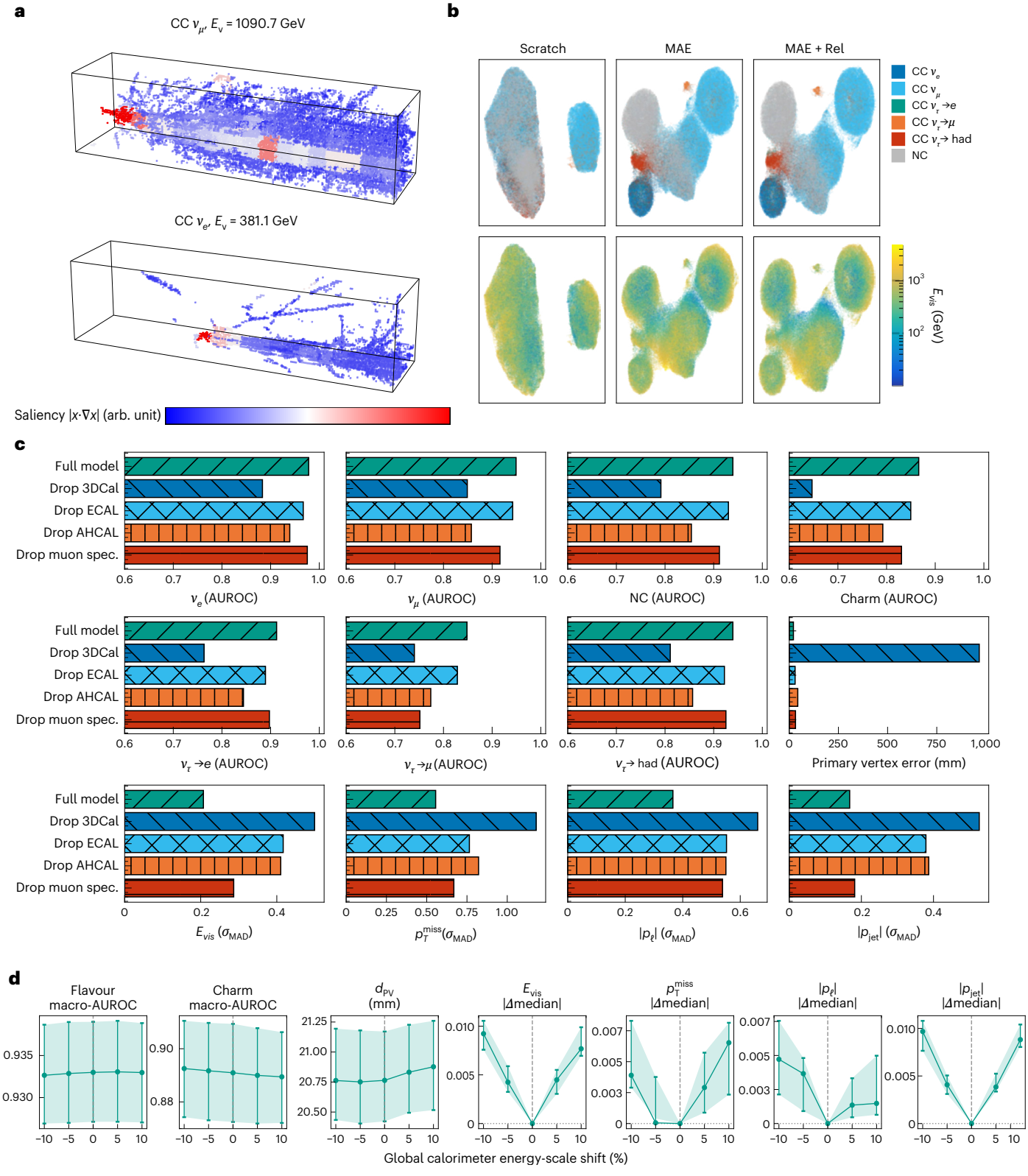
Data efficiency study

Figure 4 isolates one of the most practically important effects of pretraining: reduced dependence on large labelled training sets. The study compares Scratch, MAE and MAE + Rel across label budgets from 10^2 to 10^5 training events while keeping the validation and test sets fixed. Each point is averaged across three random seeds.

The completed three-way comparison shows that both pretrained initializations generally outperform Scratch, with MAE + Rel giving the strongest overall performance across the study. Masked reconstruction accounts for a substantial part of the data-efficiency gain, while the relational continuation provides a further, task-dependent improvement. At 10^3 labelled events, flavour macro-AUROC rises from 0.736 for Scratch to 0.790 for MAE and 0.818 for MAE + Rel; the full pretrained model already exceeds the 0.807 reached by Scratch at 10^4 events. Charm macro-AUROC at the same budget rises from 0.548 to 0.581 and 0.593, respectively, with MAE + Rel matching Scratch trained on



approximately three times more events. The classification advantage persists at 10^5 events, where MAE + Rel remains the best-performing initialization for both flavour and charm classification.

The regression results show the same broad ordering, although with more variation between targets and label budgets. At 10^3 events, the mean vertex error falls from 239 mm for Scratch to 107 mm for MAE and 100 mm for MAE + Rel. MAE + Rel also gives the clearest sustained advantage for missing transverse momentum and is generally the strongest for vertex reconstruction. For jet momentum, however, σ_{MAD} at 10^3 events falls from 0.663 to 0.454 and 0.517, so MAE alone is the strongest at this budget; MAE-only is likewise competitive or superior at several budgets for visible-energy and primary-lepton-momentum regression. At the smallest budgets of 100–300 events, pretraining is not consistently advantageous across all continuous targets. Thus, the full relational objective is generally superior across the task suite, particularly for classification, vertexing and missing transverse momentum, but its advantage is not universal at every budget or for every regression target.

Alternative-generator robustness and cross-domain transfer

The broader motivation for pretraining is not only improved downstream performance within one simulated detector domain but also the possibility of reusing representations across modelling choices, detector geometries, sensing modalities and energy regimes. We therefore collect the generator-stress and transfer results in a single summary table. Table 1a presents a test of generator dependence with a matched GENIE-to-NuWro comparison in the source detector. Table 1b,c presents a test of cross-domain transfer to a fine-grained plastic-scintillator benchmark and to PILArNet, respectively.

For the alternative-generator test, the same three fine-tuned checkpoints are evaluated without retraining on a NuWro[38] sample matched one-to-one to the GENIE test sample at the level of the incoming neutrino, interaction vertex and target nucleus. The detector simulation and reconstruction inputs are kept in the same format as for the main evaluation. Further details on data generation are given in the 'Matched alternative-generator sample' section in the Methods.

Table 1a presents a deliberately mixed, and therefore useful, outcome. The directly evaluated GENIE results preserve the in-domain ordering: Scratch is the weakest overall, MAE recovers most of the flavour and kinematic performance, and MAE + Rel is marginally the strongest on the classification metrics and on most reported regression quantities. The flavour macro-AUROC values are 0.893, 0.929 and 0.932 for Scratch, MAE and MAE + Rel, respectively; the corresponding charm macro-AUROCs are 0.790, 0.863 and 0.864. All three NuWro rows are direct evaluations of the corresponding fine-tuned checkpoints. For MAE + Rel, flavour macro-AUROC changes only mildly from 0.932 to 0.928, and the kinematic resolutions and median vertex error likewise show small changes. By contrast, charm macro-AUROC decreases from 0.864 to 0.721. The underlying one-versus-rest charm AUCs for no-charm, charm → e, charm → μ and charm → had change from (0.877, 0.809, 0.891 and 0.877) to (0.727, 0.681, 0.755 and 0.719). Taken together, the small measured shifts in flavour identification, kinematic reconstruction and vertex localization are encouraging within this matched generator comparison but do not establish generator independence. Charm tagging shows substantial sensitivity to generator-dependent heavy-flavour production, hadronization and event topology.

Table 1b,c presents a test of whether the source-domain encoder carries useful information outside the original detector and also a clarification of the role of MAE-only pretraining in transfer. The scintillator benchmark of ref. 39 is technologically close to 3DCAL, because both detectors use a fine-grained plastic scintillator, but differs in detector dimensions, magnetic environment, task definition and energy scale. Relative to target-domain training from random initialization, the MAE row already improves the diagonal entries for all four classes: from 0.919 to 0.934 for protons, from 0.532 to 0.584 for charged pions, from 0.661 to 0.723 for muons and from 0.712 to 0.768 for electrons. The full MAE + Rel then increases them further to 0.943, 0.609, 0.748 and 0.787. This indicates that masked reconstruction accounts for a large fraction of the transferred geometric and calorimetric prior, while the relational objective provides an additional gain, especially for the pion, muon and electron classes.

PILArNet[40] introduces a stronger shift, from scintillating voxels to liquid-argon time-projection-chamber data and from neutrino-event analysis to particle-level classification. In the single-particle task, the MAE result improves over Scratch from 0.8798 to 0.9050 in accuracy and from 0.834 to 0.875 in entropy-based AUROC; MAE + Rel further reaches 0.9154 and 0.891. The single-particle comparison with published values should be read carefully because of the public-resolution caveat in Table 1. The multi-particle task is directly comparable on the public PILArNet benchmark: the MAE row improves over Scratch from 0.9333 to 0.9580 in accuracy and from 0.922 to 0.944 in AUROC, while MAE + Rel reaches 0.9662 and 0.951, edging past the best published values in both metrics. Taken together, the two transfer studies suggest that masked reconstruction already learns reusable detector geometry and that the relational objectives further sharpen the representation when the target task demands particle-level or multi-particle discrimination (Fig. 5).

Discussion

The main finding of this study is that self-supervised pretraining becomes the most valuable precisely where energy-frontier neutrino events are the hardest to interpret. A single pretrained encoder improves flavour classification, charmed-quark identification, kinematic regression and vertex reconstruction, and the gains are largest in channels where shower overlap, secondary activity and partial containment make the event topology genuinely ambiguous. This matters because these channels carry direct physics relevance: they are closely tied to the tau-neutrino and heavy-flavour measurements that motivate forward collider-neutrino programmes. In this regime, representation learning is not simply refining an already mature reconstruction chain but it is helping to establish a viable analysis strategy.

The comparison between MAE and MAE + Rel also clarifies what the pretraining objective is contributing. Masked reconstruction alone already captures a substantial fraction of the global shower geometry and cross-detector context, as reflected in the improvements for the dominant flavour categories and in the broadly better regression behaviour with respect to Scratch. The relational targets add a more local and semantic constraint, and their benefit is the most visible in the lower-yield tau and charmed-quark channels, where ghost suppression, secondary structure and local ambiguities matter the most. This interpretation is now supported by three-way comparisons in the in-domain, data-efficiency and cross-domain transfer studies. Across label budgets, MAE alone accounts for most of the improvement over Scratch, while MAE + Rel provides a consistent additional classification gain and selected improvements, notably for missing transverse momentum and vertex reconstruction, without uniformly improving every regression target. Recent work in detector and neutrino machine learning has similarly shown that self-supervized objectives can improve robustness or transferability through contrastive learning, masked point modelling and related pretext tasks[33–35]. Related studies in broader particle physics have reported transferable gains from self-supervised pretraining and fine-tuning, although mostly in collider-jet or analysis-level settings rather than heterogeneous detector reconstruction[41–44]. What distinguishes the present setting is the combination of severe event complexity, heterogeneous detector inputs and joint downstream inference. The present results therefore suggest that, for dense neutrino events, reconstruction-style objectives are a strong starting point but not the whole answer: physics-aware

https://doi.org/10.1038/s42256-026-01309-6

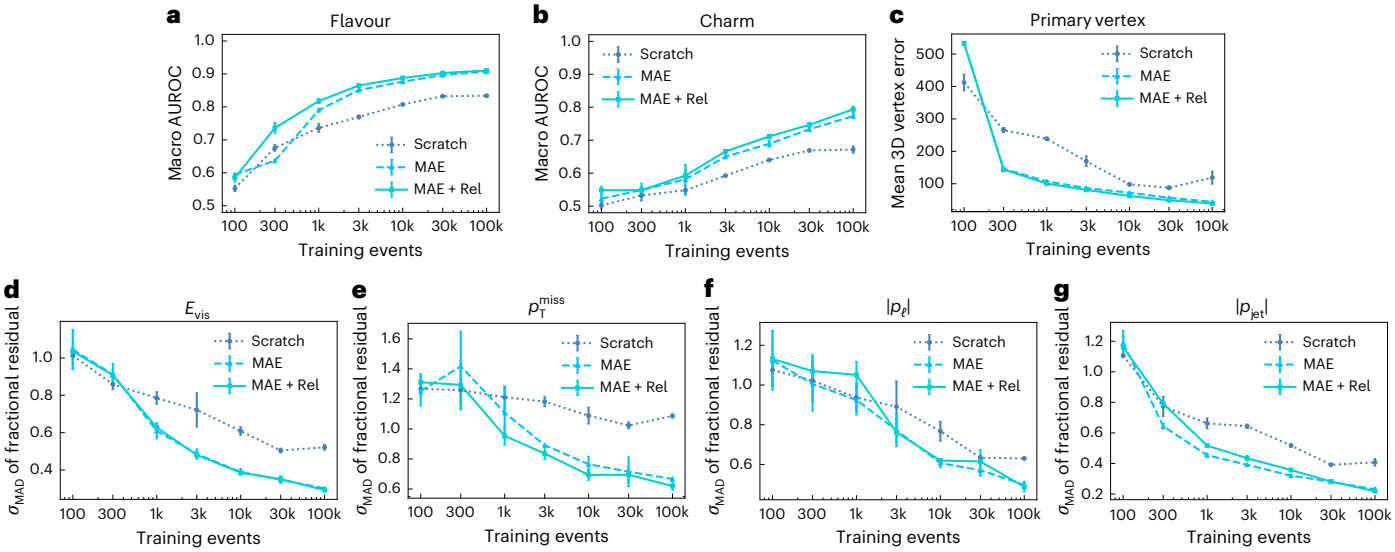


Fig. 4 | Data-efficiency study for the three initialization strategies. Performance of Scratch, MAE and MAE + Rel across labelled training budgets from 100 to 100,000 events. **a**,**b**, Flavour and charm macro-AUROC. **c**, mean 3D primary-vertex error. **d**–**g**, σ_{MAD} of the fractional residuals for visible energy, missing transverse momentum, primary-lepton momentum and jet momentum, respectively. Points are means across three random seeds and error bars show their standard deviation. Higher values are better for AUROC; lower values are better for regression and vertex errors.

local supervision can further sharpen the latent representation where the ambiguity is the greatest.

The attribution and representation diagnostics are consistent with that picture. The saliency examples emphasize the interaction region and the main downstream structures rather than diffuse occupancy alone. The latent-space UMAP maps show a more orderly geometry under MAE + Rel, with clearer flavour grouping and smoother visible-energy variation than Scratch. Detector-branch ablations then recover a physically sensible division of labour: 3DCAL carries the backbone of the inference, AHCAL contributes the most strongly to hadronic and NC discrimination, the muon spectrometer to muonic signatures and ECAL to energy-related observables. A complementary coherent energy-scale stress test produces only mild drifts over a ±10% scan, with nuisance-induced bias changes at the level of a few 10^{-3} to 10^{-2} in the current evaluation, suggesting that the representation is not overly sensitive to global calorimetric miscalibration. These analyses do not by themselves explain the model exhaustively but they strengthen the case that the learned representation is structured rather than ad hoc. They should therefore be viewed as physically motivated consistency checks, not as proof that the network has learned a fully interpretable event-reconstruction procedure.

The data-efficiency study strengthens that interpretation. Other recent detector studies have reported that self-supervised pretraining can reduce downstream label requirements[35,44,45], but those results were obtained in simpler settings, such as individual particle trajectories or detector pulse waveforms. Here, the effect is large enough to be operationally meaningful in a heterogeneous, multi-task neutrino problem. With roughly 10^3 labelled events, MAE + Rel reaches flavour-classification performance exceeding Scratch trained on roughly 10^4 events, while MAE already provides most of the improvement in jet regression and vertex reconstruction. The separation of the two pretrained variants shows that the reduction in label requirements is principally a masked-reconstruction effect, supplemented by relational gains that are the strongest for classification and selected continuous targets. That leftward shift of the performance frontier matters because, in an energy-frontier neutrino programme, labels are not a cheap by-product. Less abundant channels require dedicated simulation campaigns, systematic variations are costly and many target quantities depend on expensive truth association or high-level reconstruction. A representation that reduces the demand for supervision therefore changes not only how efficiently models can be trained but also which physics studies are practical to pursue.

The transfer results place the work in a broader context. Transfer learning in neutrino physics has so far been used mainly to adapt image-based backbones to downstream classification tasks or to reduce domain mismatch between nearby source and target problems[29,30,36]. Representation-learning studies have also begun to assess whether pretrained encoders can improve robustness within a detector family[33], while broader particle-physics work has started to test transfer from pretrained encoders across tasks and datasets in collider settings[43,44]. The transfer experiments here are deliberately staged. We first move to a public scintillator detector that is close to 3DCAL in sensing modality and voxel size but different energy regime, then to the public PILArNet benchmark, which introduces a different detector technology and a different energy regime too. In the scintillator case, transfer improves the class-conditional confusion-matrix diagonals for all four particle species and surpasses the strongest published baseline for protons, muons and electrons while approaching it for pions. In PILArNet, a source encoder pretrained on tera-electronvolt-scale neutrino interactions adapts to a public LArTPC particle-identification benchmark, improves both single-particle and multi-particle performance over scratch training and on the multi-particle task also edges past the strongest published ensemble baseline[46]. Taken together, these results suggest that the pretraining captures structure that is useful well beyond the original detector, task and energy range.

The conclusions should be read with clear limits in mind. The study is simulation based, the pretraining evidence is mainly qualitative and the relational targets rely on simulation truth. Those targets are useful precisely because they encode local detector semantics, but they may also inherit generator and detector-model assumptions. The matched NuWro evaluation in the 'Alternative-generator robustness and cross-domain transfer' section in the 'Results' adds a controlled check of this risk. The close agreement in flavour identification, kinematic reconstruction and vertex localization provides encouraging evidence that the learned representation is robust to generator-specific differences for these tasks within the matched configuration. Charm tagging sets an important boundary: its substantial degradation indicates sensitivity to neutrino interaction and heavy-flavour production,

https://doi.org/10.1038/s42256-026-01309-6

Table 1 | Generator robustness and cross-domain transfer benchmarks

(a) Matched GENIE-to-NuWro stress test

Model	Flavour AUC	Charm AUC	E_{vis}	p_T^{miss}	$\lvert p_\ell \rvert$	$\lvert p_{jet} \rvert$	d_{PV} [mm]
Scratch	0.893/0.889	0.790/0.647	0.386/0.390	0.705/0.721	0.447/0.448	0.264/0.274	22.1/22.3
MAE	0.929/0.925	0.863/0.720	0.256/0.261	0.535/0.552	0.395/0.397	0.190/0.199	16.9/17.1
MAE+Rel	0.932/0.928	0.864/0.721	0.254/0.258	0.543/0.552	0.396/0.398	0.187/0.196	16.1/16.3

(b) Transfer to the fine-grained plastic-scintillator benchmark

True class	Method	Predicted P	Predicted $\pi^\pm$	Predicted $\mu^\pm$	Predicted $e^\pm$
P	GBDT-Transformer[39]	0.907	0.067	0.007	0.019
	GBDT-RNN[39]	0.896	0.073	0.006	0.025
	GBDT-SIR-PF[39]	0.891	0.077	0.008	0.024
	Scratch (ours)	0.919	0.060	0.003	0.018
	MAE (ours)	0.934	0.049	0.005	0.012
	MAE+Rel (ours)	**0.943**	0.040	0.006	0.011
$\pi^\pm$	GBDT-Transformer[39]	0.057	**0.643**	0.041	0.259
	GBDT-RNN[39]	0.080	0.623	0.036	0.261
	GBDT-SIR-PF[39]	0.080	0.606	0.042	0.272
	Scratch (ours)	0.062	0.532	0.267	0.139
	MAE (ours)	0.056	0.584	0.244	0.116
	MAE+Rel (ours)	0.053	0.609	0.227	0.111
$\mu^\pm$	GBDT-Transformer[39]	0.071	0.190	0.595	0.144
	GBDT-RNN[39]	0.089	0.233	0.506	0.172
	GBDT-SIR-PF[39]	0.126	0.236	0.517	0.121
	Scratch (ours)	0.020	0.173	0.661	0.146
	MAE (ours)	0.020	0.108	0.723	0.149
	MAE+Rel (ours)	0.019	0.079	**0.748**	0.155
$e^\pm$	GBDT-Transformer[39]	0.020	0.199	0.009	0.772
	GBDT-RNN[39]	0.027	0.200	0.007	0.766
	GBDT-SIR-PF[39]	0.017	0.237	0.006	0.740
	Scratch (ours)	0.019	0.179	0.091	0.712
	MAE (ours)	0.015	0.108	0.109	0.768
	MAE+Rel (ours)	0.012	0.084	0.117	**0.787**

(c) Transfer to PILArNet particle-identification benchmarks

Method	Single-particle classification*		Multi-particle classification	
	Accuracy	AUROC	Accuracy	AUROC
Deterministic[46]	0.8656	0.753	0.9604	0.938
Naive Ensembles[46]	0.8844	0.827	0.9640	0.944
Bootstrap Ensembles[46]	0.9014	0.842	0.9644	0.942
MC Dropout[46]	0.8734	0.795	–	–
EDL-MLL[46]	0.8622	0.762	0.9604	0.935
EDL-BR[46]	0.8253	0.701	0.9223	0.900
EDL-Brier[46]	0.8751	0.748	0.9596	0.911
Scratch (ours)	0.8798	0.834	0.9333	0.922
MAE (ours)	0.9050	0.875	0.9580	0.944
MAE+Rel (ours)	**0.9154**	**0.891**	**0.9662**	**0.951**

A single table collects the three external-validation summaries. Panel **a** reports directly evaluated matched GENIE/NuWro performance in the source detector for all three models, with entries shown as GENIE/NuWro pairs. The classification columns are macro-averaged one-versus-rest AUROC values; the regression columns report σ_{MAD} of fractional residuals, except d_{PV}, which is the median primary-vertex error. Panel **b** gives the full row-normalized confusion matrix for the fine-grained plastic-scintillator benchmark; published baselines are taken from Table 3 of ref. 39, and rows sum to 1. Panel **c** gives the full PILArNet comparison; all published baseline rows are from ref. 46. The single-particle comparison marked with * is not strictly like-for-like because the reference paper reports the published values on a same-generation $1{,}024^3$ dataset, whereas our rows use the public 768^3-pixel release. The MAE rows in panels **b** and **c** report the final MAE-only transfer results. Bold numerical entries mark the best-performing value among directly comparable methods: the largest diagonal entry for each true class in panel **b** and the best metric value in panel **c**.

hadronization and topology. We therefore interpret the NuWro comparison as evidence of task-dependent generator robustness, not generator independence or complete systematic validation. Broader generator reweighting, flux variations, detector-response mismodelling and experimental-data calibration remain necessary before deployment. Recent work on masked modelling and domain adaptation in neutrino experiments has highlighted the importance of testing whether learned representations remain stable under simulation mismodelling and domain shift[30,34,36]. Stronger claims about broadly reusable detector models will require wider transfer studies, dedicated stress tests and, ultimately, validation on experimental data.

A related deployment question is how errors in one detector branch propagate through a shared multimodal latent representation. The ablation study shows that the branches contribute in channel-dependent ways, but suppressing a branch is not the same as modelling a biased or inefficient branch. A realistic deployment programme should therefore include subsystem-specific perturbations, such as independent calorimeter-scale shifts, dead or inefficient channels, altered thresholds and noise, muon-spectrometer inefficiencies and detector-response mismodelling, and should examine whether such perturbations bias all downstream tasks through the fused representation. Uncertainty quantification is also essential for experimental use. Possible extensions include nuisance-aware training, calibrated ensembles, Bayesian or evidential task heads, conformal prediction and latent-space or reconstruction-based out-of-distribution scores. Such tools could make the learned representation useful not only for supervized reconstruction but also for rare-event triage or discovery-oriented searches. The present work does not demonstrate anomaly detection or unseen-physics discovery; it instead establishes the representation-learning basis on which such studies could be built.

We therefore do not claim that a general detector foundation model has been achieved. Rather, the results show that several ingredients usually discussed separately can coexist in one detector-aware sparse encoder: self-supervised pretraining, joint downstream fine-tuning, meaningful low-label gains and non-trivial transfer beyond the source detector. For energy-frontier neutrino physics, where event complexity makes artificial intelligence a requirement rather than an optional convenience, that is already a substantive result. It also outlines a concrete path towards more general detector encoders and motivates further work on hybrid pretraining objectives and domain adaptation.

Methods

Case-study detector and simulated data

We use the proposed FASERCAL concept at the Large Hadron Collider at CERN as a representative energy-frontier case study[22]. In this setup, neutrino interactions occur inside the highly granular 3DCAL, a ten-module detector with 48 × 48 × 200 scintillator voxels in total, which is followed downstream by the electromagnetic calorimeter (ECAL), the hadronic calorimeter (AHCAL) and a muon spectrometer. We refer to the primary calorimeter as 3DCAL throughout, reserving FASERCAL for the broader detector concept. An example event spanning the detector subsystems is shown in Supplementary Fig. 3.

Run-4 forward neutrino fluxes from light-hadron and charm-hadron sources are used to generate interactions with GENIE v3.04.00[47], tau-lepton and charm-hadron decays are modelled with PYTHIA8[48] and the resulting particles are propagated through the full detector geometry with Geant4[22,49]. This matters physically because the flavour composition and energy spectrum depend on the parent-hadron origin: charm dominates the ν_τ component and contributes substantially to the high-energy ν_e flux in the far-forward region. The collaboration simulation and reconstruction framework underlying these studies is publicly available at https://github.com/rubbiaa/FASER.

The nominal simulated sample corresponds to 101 ab^{-1} of integrated luminosity and yields 1,118,058 neutrino interactions in the 3DCAL modules. To improve training statistics for the less abundant and most topologically difficult channels, we add 108,317 dedicated ν_τ CC interactions from an enriched sample corresponding to 3,700 ab^{-1}. The combined event pool is split once into train, validation and test partitions of 85/5/10. As this pooled set over-represents ν_τ events, all the results presented in this Article are reweighted at evaluation time so the ν_τ abundance matches the nominal unbiased sample; this is the procedure implemented in the analysis code used to generate the paper figures.

The detector inputs are heterogeneous by construction. The 3DCAL provides sparse 3D voxel hits with charge information and simulation-derived voxel labels used only during pretraining. The AHCAL provides a second sparse calorimetric volume at coarser granularity. The ECAL is represented as a compact 5 × 5 energy matrix, and the muon spectrometer provides up to ten hit-measuring planes per track. This mixture of sparse volumetric inputs, dense global summaries and variable-length track information is precisely why a unified encoder is non-trivial. Before tokenization, the 3DCAL and AHCAL inputs are treated as sparse lists of occupied voxel coordinates and charge-like readout values, with charge transformed using the same logarithmic preprocessing used throughout training. The ECAL input is a dense two-dimensional energy summary, encoded as a single detector token after a small learned projection. The muon-spectrometer input is variable length: track or hit-plane summaries are embedded and pooled into a compact spectrometer token, together with presence/count information. Figure 5b shows how these four native detector representations are converted into detector-specific tokens and fused, while Supplementary Fig. 3 shows a representative event spanning the same four subsystems. Read together, the two figures connect the physical detector response to the precise inputs and preprocessing supplied to the model before multimodal fusion.

Simulation also provides auxiliary truth that is useful during pretraining. In addition to event-level labels for the downstream tasks, we exploit voxel-level occupancy, charge, ghost labels, hierarchy labels and particle-category labels inside the 3DCAL. Ghost labels identify reconstructed voxels with no matched true particle. Hierarchy labels separate background activity from voxels dominated by primary or secondary particles, while particle-category labels group matched truth deposits into electromagnetic, muonic and hadronic activity. The ghost target is binary, but the two semantic targets are not hard one-hot labels: a reconstructed voxel can be matched to several true deposits, so hierarchy and particle-category supervision are represented as soft per-voxel distributions obtained by summing the matched fractional contribution weights for each class and normalizing within the voxel. This formulation is necessary because dense shower regions often mix contributions from several particles within a single reconstructed voxel.

Matched alternative-generator sample

To probe the dependence of the trained models on the neutrino-interaction generator, we constructed a NuWro[38] sample matched event-by-event to the GENIE test set. For each GENIE test interaction, we generated a single NuWro interaction with the same incoming-neutrino four-momentum (energy, flavour and direction), the same struck target nucleus and the same current (charged or neutral); the primary-interaction vertex was fixed to the GENIE value. Given this initial state, NuWro selected the exclusive reaction channel and produced the hadronic final state according to its own cross-section, nuclear and hadronization models, with deep-inelastic hadronization handled by PYTHIA6[50]. The matching holds the flux, detector geometry and target composition fixed, making the comparison primarily sensitive to differences in the primary-interaction and hadronization models rather than to exposure or acceptance.

Tau leptons and charmed hadrons were propagated undecayed out of the primary generator and decayed by a common downstream

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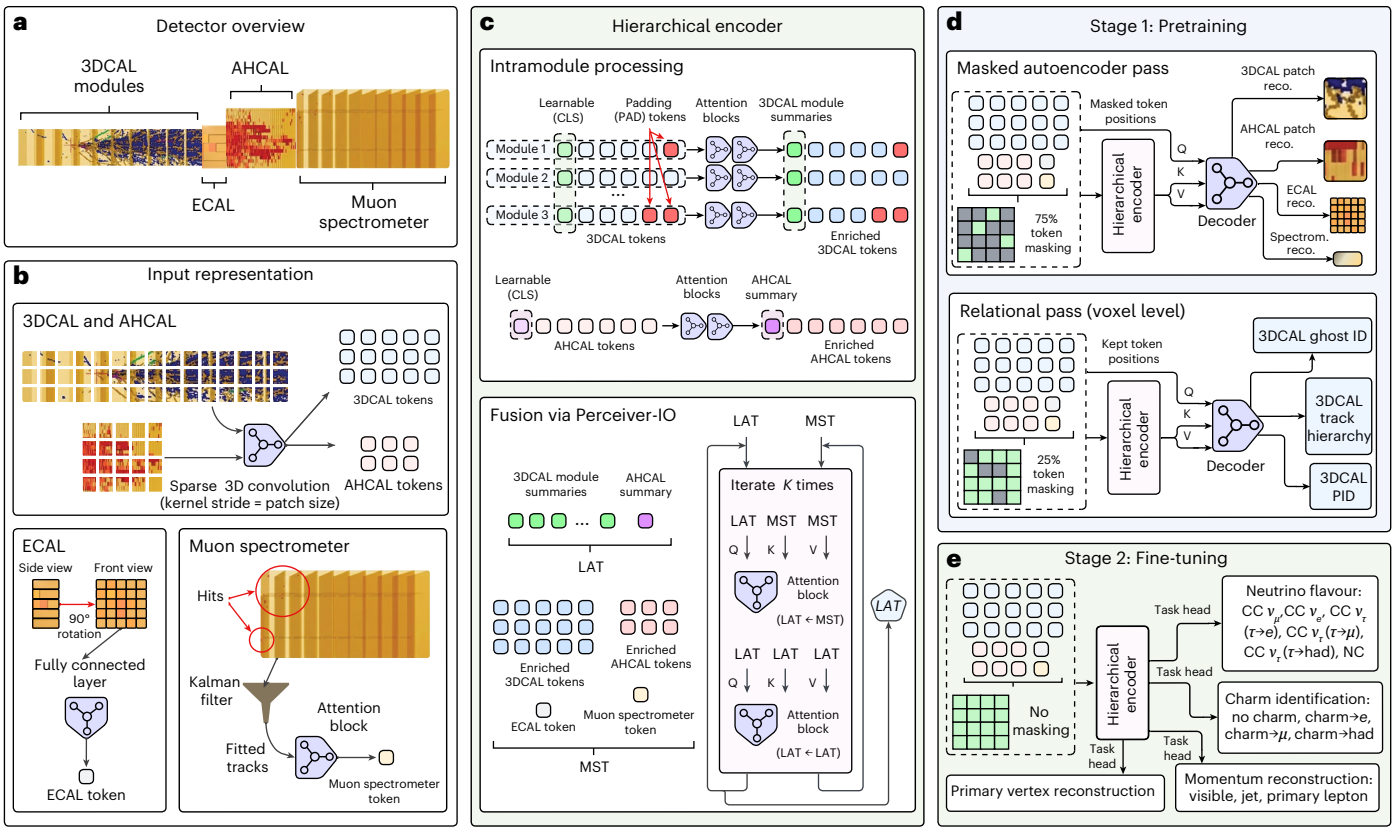

Fig. 5 | Overview of the framework. a, FASERCAL case study, with the 3DCAL as the primary interaction volume followed by the ECAL, AHCAL and muon spectrometer. **b**, The sparse 3DCAL and AHCAL voxel inputs, dense 5 × 5 ECAL energy matrix and variable-length muon-spectrometer summaries are converted into detector-specific tokens. **c**, A hierarchical encoder performs module-level self-attention before Perceiver-IO fusion of the heterogeneous detector streams, where Q, K, and V denote query, key and value representations, respectively. LAT, latent; MST, multi-source token. **d**, Stage 1 pretraining combines masked reconstruction with a relational voxel-level pass. reco., reconstruction. **e**, The pretrained encoder is then fine-tuned jointly on event-level classification and regression tasks. A representative event spanning the four input subsystems is shown in Supplementary Fig. 3.

stage based on PYTHIA8[48], identical to the one used in the nominal GENIE production. The τ and heavy-flavour decay modelling is therefore shared between the two samples, and the $\tau \to e/\mu$/had and charm → e/μ/had categories are defined from the decay products exactly as in Sec. II B. Charm production in NuWro proceeds through the slow-rescaling mechanism; the resulting charmed CC fraction is 9.2%, compared with 9.66% in the GENIE sample, and the simulated τ-decay branching fractions are consistent with their Standard-Model values. Because the τ leptons and charmed hadrons produced by the two generators carry different kinematics, their decays are independent realizations of a common decay model rather than event-by-event replicas; only the incoming neutrino state, vertex and target nucleus are matched.

Neutrino–electron elastic-scattering events have no nuclear-target counterpart in this matched configuration and were not reproduced; they constitute a negligible fraction of the sample.

To preserve ν_τ statistics, the matching reproduces the full composition of the GENIE test set, including its τ-enhanced beam component. Each matched event was passed through the same FASERCAL Geant4 detector simulation and reconstruction chain as the nominal sample, producing identical input formats. The final evaluated NuWro sample contains 121,371 events.

Sparse multimodal encoder

All headline experiments use the same base encoder. The model follows the design shown in Fig. 5. Sparse 3D convolutions use the SpConv framework[51] to convert the 3DCAL and AHCAL voxel grids into patch tokens while operating only on occupied regions[52]. The 3DCAL is tokenized in patches of 12 × 12 × 10 voxels, yielding a 4 × 4 × 20 patch grid of up to 320 tokens, of which only those overlapping at least one active voxel are retained. The AHCAL is tokenized in patches of 6 × 6 × 5 voxels (3 × 3 × 8 grid, up to 72 tokens). This sparse tokenization makes the computational cost scale with detector occupancy rather than with the total instrumented volume.

The token sequence is then processed in two stages. First, 3DCAL tokens are grouped by detector module (the detector comprises ten longitudinal modules, each spanning two patch planes in depth) and processed through module-level self-attention blocks augmented with learned module-position embeddings and per-module class tokens. This structure allows the model to capture local shower patterns within each module before mixing information across the detector. The AHCAL branch is processed through a parallel self-attention stack with its own class tokens. All attention layers use an embedding dimension of 384 with 12 heads (head dimension of 32) and a multi-layer perceptron ratio of 4.

Second, a Perceiver-IO bottleneck[53] fuses the calorimetric tokens with compact representations of the ECAL (encoded as a single token from its 5 × 5 energy matrix) and muon spectrometer (encoded from up to ten hit-measuring planes per track). Learned detector-type embeddings distinguish the sources. The cross-attention and self-attention layers in the Perceiver loop produce a fixed-size latent representation that can be consumed by either the pretraining decoder or the downstream task heads.

The downstream fine-tuning and inference model used in the headline experiments contains 31,936,341 trainable model parameters; including the six learned homoscedastic loss-weight scalars used

only during optimization gives 31,936,347 optimized scalars during fine-tuning. This hierarchy is important for two reasons. Computationally, it avoids global attention over an unnecessarily large token sequence. Scientifically, it respects the detector's physical organization while still allowing global event context to emerge in the latent space. The architecture therefore provides a natural compromise between local geometric fidelity and cross-detector integration. When loading pretrained weights into the fine-tuning model, matching layers are transferred and the additional layers are randomly initialized. Additional architectural details are shown in Supplementary Fig. 3 and presented in Supplementary Table 1.

Self-supervised pretraining (stage 1)

Pretraining combines two complementary objectives applied in a two-phase schedule. In both phases, the core objective is masked reconstruction in the spirit of MAE[24]: 75% of occupied calorimeter patches are randomly masked, and a lightweight decoder (embedding dimension 256, eight heads) cross-attends from mask-token queries to the encoder's latent representation to predict voxel-level occupancy and charge in the missing regions. To map a single decoder token back to the many voxels within its patch, a multi-rank separable basis (initialized from discrete cosine transform coefficients) is used. Auxiliary detector inputs (ECAL and muon spectrometer) are also masked and reconstructed when applicable. The masking is occupancy aware, so the model concentrates capacity on non-trivial targets.

In the first phase, the model is trained for 400 epochs using masked reconstruction alone, allowing the encoder to learn broad spatial correlations and cross-detector context. The checkpoint at the end of this phase defines the MAE encoder used in the downstream comparisons. In the second phase, training continues from this checkpoint for 100 additional epochs while introducing, with probability 0.5 per batch, a relational forward pass in which the encoder predicts voxel-level ghost labels, hierarchy labels and particle-category labels on kept 3DCAL patches (with a lower mask ratio of 0.25). Ghost remains a hard binary target, whereas hierarchy and particle-category supervision use soft labels built from the normalized fractional contributions of all matched truth deposits in a voxel. The semantic classes are therefore allowed to overlap at voxel level, which makes the task particularly demanding in dense shower regions. The checkpoint at the end of the second phase defines the MAE + Rel encoder. The two-phase schedule avoids premature saturation on the relational tasks, which are inherently easier than masked reconstruction. The MAE + Rel continuation required 427 graphics processing unit (GPU)-hours in addition to the 584 GPU-hours used for MAE pretraining, an incremental pretraining cost of 73%. The relational decoder and heads are discarded before fine-tuning, so this continuation does not change the downstream parameter count or inference path. The incremental gains over MAE alone are modest for many aggregate AUROC values but are larger for selected low-yield operating points; for example, the maximum figure of merit for $\nu_\tau \mathrm{CC} \to e$ increases from 1.03 to 1.35. The full objective therefore trades substantial additional one-off pretraining cost for targeted gains in several difficult channels, rather than a uniform improvement across all observables.

All pretraining losses are combined using learned homoscedastic uncertainty weights[54]. The reconstruction losses use a hybrid formulation that includes soft-chamfer style[55] and distance-weighted regression terms, so that predictions close to the true sparse support are penalized more gently than those that miss the relevant structure entirely. The optimiser is AdamW[56] with a base learning rate of 10^{-4} (linearly scaled by effective batch size), $\beta_1 = 0.9$, $\beta_2 = 0.95$ and weight decay 0.05. Phase 1 uses 40 warm-up epochs and cosine annealing over 360 epochs; the shorter phase 2 continuation uses 10 warm-up epochs and cosine annealing over 90 epochs. Pretraining is distributed across two nodes of four NVIDIA GH200 GPUs each (eight GPUs total) with a per-GPU batch size of 512. Training hyperparameters and active compute costs are presented in Supplementary Tables 1 and 2.

Joint fine-tuning (stage 2)

For downstream learning, the MAE decoder is discarded and the shared encoder is retained. Task-specific tokens or lightweight heads read out the latent representation and jointly predict neutrino flavour, charmed-quark category, visible momentum, jet momentum and the primary interaction vertex. Primary-lepton momentum is derived from the reconstructed event components and is used in the evaluation plots. For a fair comparison, all downstream variants use the same fine-tuning architecture and the same random initialization for task-specific heads; only the encoder initialization differs between Scratch, MAE and MAE + Rel.

Fine-tuning is multi-task throughout. This matters because the downstream observables are physically coupled: flavour identification, charm production, visible energy flow and vertex position all depend on a common interpretation of the same event. Joint fine-tuning therefore tests whether the pretrained representation is useful as a shared basis for multiple measurements rather than for a single optimized classifier.

Transfer-learning targets and adaptation

Transfer experiments retain the transferable core of the source encoder while replacing detector-specific components. In practice, the attention blocks, latent cross-attention blocks, latent self-attention blocks, normalization layers and global query token are inherited when shapes and semantics match, whereas detector-specific patch embeddings, positional encodings, detector-branch embeddings and task heads are reinitialized for the target domain. Scratch baselines use the same target architectures without transferred weights.

One target is the public fine-grained plastic-scintillator benchmark associated with ref. 39. This dataset provides four-class particle identification on isolated charged-particle tracks at giga-electronvolt (GeV) scale energies in a 3D detector built from 1-cm^3 scintillator elements. We train on the provided public training split, reserve 5% of it for validation and evaluate on the provided testing split. The model operates on local $120 \times 120 \times 120$ voxel crops extracted from the native 200^3 detector and receives a compact context token summarizing crop geometry and detector-boundary position, both derived deterministically from the same voxel hits. This preserves containment information without relying on fitted trajectories or external reconstruction.

The second target is PILArNet[40]. Here, the detector technology changes to a LArTPC, and the downstream tasks are five-class single-particle and multi-particle classification. We use the public 768^3-pixel release throughout and construct a fixed 80k/2k/18k train/validation/test event split from the public HDF5 files. Single-particle classification uses centred particle crops with a compact voxel-derived metadata branch. Multi-particle classification adds a lightweight context transformer over the shared transferred encoder to aggregate particles from the same event, together with voxel-derived per-particle context features. These metadata branches are included to compensate for cropping or rescaling, not to inject external detector information. For fair comparison with published literature, transfer performance is reported with micro-accuracy and the entropy-based AUROC definition used in ref. 46.

Evaluation protocol

The main in-domain classification and regression comparisons use all three downstream variants defined at the start of the 'Results'. Studies that currently compare only Scratch and MAE + Rel are interpreted as tests of the full pretraining recipe relative to random initialization, not as measurements of the relational contribution. Classification performance is reported with one-versus-rest operating curves and confusion matrices, together with threshold scans based on the figure of merit $\mathrm{FOM} = S/\sqrt{S+B}$, where S and B denote the expected signal and background yields after selection. Regression performance is summarized through selected-sample residual distributions and

vertex-error distributions. Transfer benchmarks use fixed public splits, with the scintillator validation hold-out and the PILArNet manifests defined above.

The data-efficiency study keeps the validation and test sets fixed, varies only the number of labelled training events and repeats each setting across three random seeds. This isolates the extent to which pretraining reduces labelled-data requirements. Exact manifests and checkpoint-selection rules are provided in the released code, while the data splits and target definitions are summarized in Supplementary Information, sections 3.4 and 4.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The simulated FASERCAL dataset underlying the main results of this manuscript was produced with the publicly available FASERCAL simulation and reconstruction framework, available via GitHub at https://github.com/rubbiaa/FASER. The complete held-out FASERCAL test dataset used to evaluate the models and obtain the principal results reported in this article is publicly available via Zenodo at https://doi.org/10.5281/zenodo.22011619 (ref. 57). The full FASERCal training and validation datasets are approximately 500 GB and, owing to their large storage requirements, are available from the corresponding authors on request. The datasets used for the transfer-learning studies can be found at the following links: the fine-grained plastic-scintillator benchmark is available via Zenodo at https://doi.org/10.1088/1748-0221/18/12/P12013 (ref. 46); the PILArNet data used for the LArTPC transfer study are available via the public PILArNet project at https://osf.io/bu4fp/, including the LArTPC simulation component at https://osf.io/vruzp/.

Code availability

All code used in this study is available via Zenodo at https://doi.org/10.5281/zenodo.22011299 (ref. 58) and via GitHub at https://github.com/saulam/faserDL.

References

1. FASER Collaboration. Detecting and studying high-energy collider neutrinos with FASER at the LHC. *Eur. Phys. J. C* **80**, 61 https://doi.org/10.1140/epjc/s10052-020-7631-5 (2020).
2. FASER Collaboration. First direct observation of collider neutrinos with FASER at the LHC. *Phys. Rev. Lett.* **131**, 031801 https://doi.org/10.1103/PhysRevLett.131.031801 (2023).
3. FASER Collaboration. First measurement of the ν_e and ν_μ interaction cross sections at the LHC with FASER's emulsion detector. *Phys. Rev. Lett.* **133**, 021802 https://doi.org/10.1103/PhysRevLett.133.021802 (2024).
4. Cruz-Martinez, J. M. et al. The LHC as a neutrino-ion collider. *Eur. Phys. J. C* **84**, 369 https://doi.org/10.1140/epjc/s10052-024-12665-1 (2024).
5. Mammen Abraham, R. et al. First measurement of the muon neutrino interaction cross section and flux as a function of energy at the LHC with FASER. *Phys. Rev. Lett.* **134**, 211801 (2025).
6. SND@LHC Collaboration Observation of collider muon neutrinos with the SND@LHC experiment. *Phys. Rev. Lett.* **131**, 031802 (2023).
7. Radovic, A. et al. Machine learning at the energy and intensity frontiers of particle physics. *Nature* **560**, 41–48 https://doi.org/10.1038/s41586-018-0361-2 (2018).
8. Coelho, C. N. et al. Automatic heterogeneous quantization of deep neural networks for low-latency inference on the edge for particle detectors. *Nat. Mach. Intell.* **3**, 675–686 https://doi.org/10.1038/s42256-021-00356-5 (2021).
9. Karagiorgi, G., Kasieczka, G., Kravitz, S., Nachman, B. & Shih, D. Machine learning in the search for new fundamental physics. *Nat. Rev. Phys.* **4**, 399–412 https://doi.org/10.1038/s42254-022-00441-9 (2022).
10. Govorkova, E. et al. Autoencoders on field-programmable gate arrays for real-time, unsupervised new physics detection at 40 MHz at the Large Hadron Collider. *Nat. Mach. Intell.* **4**, 154–161 https://doi.org/10.1038/s42256-022-00441-3 (2022).
11. Belis, V., Woźniak, K. A., Puljak, E. et al. Quantum anomaly detection in the latent space of proton collision events at the LHC. *Commun. Phys.* **7**, 334 https://doi.org/10.1038/s42005-024-01811-6 (2024).
12. Pata, J. et al. Improved particle-flow event reconstruction with scalable neural networks for current and future particle detectors. *Commun. Phys.* **7**, 107 https://doi.org/10.1038/s42005-024-01599-5 (2024).
13. Aurisano, A. et al. A convolutional neural network neutrino event classifier. *J. Instrum.* **11**, P09001 (2016).
14. Acciarri, R. et al. Convolutional neural networks applied to neutrino events in a liquid argon time projection chamber. *J. Instrum.* **12**, P03011 (2017).
15. Psihas, F., Groh, M., Tunnell, C. & Warburton, K. A review on machine learning for neutrino experiments. *Int. J. Modern Phys. A* **35**, 2043005 https://doi.org/10.1142/S0217751X20430058 (2020).
16. Abi, B. et al. Neutrino interaction classification with a convolutional neural network in the DUNE far detector. *Phys. Rev. D* **102**, 092003 (2020).
17. Dominé, L. & Terao, K. Scalable deep convolutional neural networks for sparse, locally dense liquid argon time projection chamber data. *Phys. Rev. D* **102**, 012005 (2020).
18. Abratenko, P. et al. Semantic segmentation with a sparse convolutional neural network for event reconstruction in MicroBooNE. *Phys. Rev. D* **103**, 052012 (2021).
19. Alonso-Monsalve, S. et al. Graph neural network for 3D classification of ambiguities and optical crosstalk in scintillator-based neutrino detectors. *Phys. Rev. D* **103**, 032005 https://doi.org/10.1103/PhysRevD.103.032005 (2021).
20. Drielsma, F. mc et al. Clustering of electromagnetic showers and particle interactions with graph neural networks in liquid argon time projection chambers. *Phys. Rev. D* **104**, 072004 https://doi.org/10.1103/PhysRevD.104.072004 (2021).
21. Alonso-Monsalve, S. et al. Deep-learning-based decomposition of overlapping-sparse images: application at the vertex of simulated neutrino interactions. *Commun. Phys.* **7**, 173 (2024).
22. Alonso-Monsalve, S. et al. *Fasercal Conceptual Design Report* https://cds.cern.ch/record/2954673 (CERN, 2026).
23. Devlin, J., Chang, M.-W., Lee, K. & Toutanova, K. BERT: pre-training of deep bidirectional transformers for language understanding. In *Proc. 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies* Vol. 1 (eds Burstein, J., Doran, C. & Solorio, T.) 4171–4186 https://aclanthology.org/N19-1423/ (Association for Computational Linguistics, 2019).
24. He, K. et al. Masked autoencoders are scalable vision learners. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (eds Dana, K., Hua, G., Roth, S., Samaras, D. & Singh, R.) 16000–16009 (IEEE, 2022).
25. Xie, Z. et al. SimMIM: a simple framework for masked image modeling. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (eds Dana, K., Hua, G., Roth, S., Samaras, D. & Singh, R.) 9653–9663 (IEEE, 2022).
26. Baevski, A. et al. data2vec: A general framework for self-supervised learning in speech, vision and language. In *Proc. 39th International Conference on Machine Learning* vol. 162 (eds Chaudhuri, K. et al.) 1298–1312 https://proceedings.mlr.press/v162/baevski22a.html (PMLR, 2022).

27. Yu, X. et al. Point-BERT: pre-training 3D point cloud transformers with masked point modeling. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (eds Dana, K., Hua, G., Roth, S., Samaras, D. & Singh, R.) 19313–19322 (IEEE, 2022).
28. Hou, Z. et al. GraphMAE: self-supervised masked graph autoencoders. In *Proc. 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '22)* 594–604 https://doi.org/10.1145/3534678.3539321 (2022).
29. Chappell, A. & Whitehead, L. H. Application of transfer learning to neutrino interaction classification. *Eur. Phys. J. C* **82**, 1099 https://doi.org/10.1140/epjc/s10052-022-11066-6 (2022).
30. Babicz, M., Alonso-Monsalve, S., Dolan, S. & Terao, K. Adversarial methods to reduce simulation bias in neutrino interaction event filtering at liquid argon time projection chambers. *Phys. Rev. D* **105**, 112009 https://doi.org/10.1103/PhysRevD.105.112009 (2022).
31. Sagar, D., Yu, K., Yankelevich, A., Bian, J. & Baldi, P. Adapting vision-language models for neutrino event classification in high-energy physics. *Commun. Phys.* **9**, 186 https://doi.org/10.1038/s42005-026-02688-3 (2026).
32. Albert, A. et al. Deep learning framework for enhanced neutrino reconstruction of single-line events in the ANTARES telescope. *Mach. Learn.: Sci. Technol.* **7**, 035004 https://doi.org/10.1088/2632-2153/ae5d84 (2026).
33. Wilkinson, A., Radev, R. & Alonso-Monsalve, S. Contrastive learning for robust representations of neutrino data. *Phys. Rev. D* **111**, 092011 https://doi.org/10.1103/PhysRevD.111.092011 (2025).
34. Yu, F., Kamp, N. & Argüelles, C. Reducing simulation dependence in neutrino telescopes with masked point modeling. In *Proc. 39th International Cosmic Ray Conference—PoS(ICRC2025)* Vol. 501, 1218 (SISSA Medialab, 2025).
35. Young, S., Jwa, Y.-j. & Terao, K. Particle trajectory representation learning with masked point modeling. *Mach. Learn. Sci. Technol.* https://doi.org/10.1088/2632-2153/ae47b8 (2026).
36. Bonilla, J. L. et al. Transfer learning for neutrino scattering: domain adaptation with generative adversarial networks. *Phys. Rev. D* **113**, 053001 https://doi.org/10.1103/kwjj-wp1c (2026).
37. Dosovitskiy, A. et al. An image is worth 16 × 16 words: transformers for image recognition at scale. In *International Conference on Learning Representations* https://openreview.net/forum?id=YicbFdNTTy (2021).
38. Golan, T., Juszczak, C. & Sobczyk, J. T. Final state interactions effects in neutrino-nucleus interactions. *Phys. Rev. C* **86**, 015505 (2012).
39. Alonso-Monsalve, S., Sgalaberna, D., Zhao, X., McGrew, C. & Rubbia, A. Artificial intelligence for improved fitting of trajectories of elementary particles in dense materials immersed in a magnetic field. *Commun. Phys.* **6**, 119 https://doi.org/10.1038/s42005-023-01239-4 (2023).
40. Adams, C., Terao, K. & Wongjirad, T. PILArNet: public dataset for particle imaging liquid argon detectors in high energy physics. Preprint at https://doi.org/10.48550/arXiv.2006.01993 (2020).
41. Harris, P., Krupa, J., Kagan, M., Maier, B. & Woodward, N. Resimulation-based self-supervised learning for pretraining physics foundation models. *Phys. Rev. D* **111**, 032010 https://doi.org/10.1103/PhysRevD.111.032010 (2025).
42. Heinrich, L. et al. Masked particle modeling on sets: towards self-supervised high energy physics foundation models. *Mach. Learn. Sci. Technol.* **5**, 035074 https://doi.org/10.1088/2632-2153/ad64a8 (2024).
43. Birk, J., Hallin, A. & Kasieczka, G. Omnijet-α: the first cross-task foundation model for particle physics. *Mach. Learn. Sci. Technol.* **5**, 035031 https://doi.org/10.1088/2632-2153/ad66ad (2024).
44. Vigl, M., Hartman, N. & Heinrich, L. Finetuning foundation models for joint analysis optimization. *Mach. Learn. Sci. Technol.* **5**, 025075 https://doi.org/10.1088/2632-2153/ad55a3 (2024).
45. Babicz, M., Alonso-Monsalve, S., Fauquex, A. & Baudis, L. Transformer-based pulse shape discrimination in HPGe detectors with masked autoencoder pre-training. *Mach. Learn.: Sci. Technol.* **7**, 045023 https://doi.org/10.1088/2632-2153/ae7ec9 (2026).
46. Koh, D. H., Mishra, A. & Terao, K. Deep neural network uncertainty quantification for LArTPC reconstruction. *J. Instrum.* **18**, P12013 https://doi.org/10.1088/1748-0221/18/12/P12013 (2023).
47. Andreopoulos, C. et al. The GENIE neutrino Monte Carlo generator. *Nucl. Instrum. Methods Phys. Res. A* **614**, 87–104 (2010).
48. Bierlich, C. et al. A comprehensive guide to the physics and usage of PYTHIA 8.3. *SciPost Phys. Codebases* **8**, r8.3 (2022).
49. Agostinelli, S. et al. Geant4: a simulation toolkit. *Nucl. Instrum. Methods Phys. Res. A* **506**, 250–303 (2003).
50. Sjöstrand, T., Mrenna, S. & Skands, P. Z. PYTHIA 6.4 physics and manual. *J. High Energy Phys.* **2006**, 026 (2006).
51. Yan, Y. SpConv: spatially sparse convolution library. *GitHub* https://github.com/traveller59/spconv (2020).
52. Graham, B., Engelcke, M. & van der Maaten, L. 3D semantic segmentation with submanifold sparse convolutional networks. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition* (eds Forsyth, D., Laptev, I., Oliva, A. & Ramanan, D.) 9224–9232 https://openaccess.thecvf.com/content_cvpr_2018/html/Graham_3D_Semantic_Segmentation_CVPR_2018_paper.html (IEEE, 2018).
53. Jaegle, A. et al. Perceiver IO: a general architecture for structured inputs & outputs. In *International Conference on Learning Representations (ICLR 2022)* https://openreview.net/forum?id=fILj7WpI-g (2022).
54. Cipolla, R., Gal, Y. & Kendall, A. Multi-task learning using uncertainty to weigh losses for scene geometry and semantics. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition* 7482–7491 (2018).
55. Fan, H., Su, H. & Guibas, L. J. A point set generation network for 3d object reconstruction from a single image. In *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition* (eds Rehg, J., Liu, Y., Wu, Y. & Taylor, C.) 605–613 https://doi.org/10.1109/CVPR.2017.72 (IEEE, 2017).
56. Loshchilov, I. & Hutter, F. Decoupled weight decay regularization. In *International Conference on Learning Representations* https://openreview.net/forum?id=Bkg6RiCqY7 (2019).
57. Alonso-Monsalve, S., Cufino, F., Kose, U., Mascellani, A. & Rubbia, A. FASERCal test data for "Towards foundation-style models for energy-frontier heterogeneous neutrino detectors via self-supervised pre-training". *Zenodo* https://doi.org/10.5281/zenodo.22011619 (2026).
58. Alonso-Monsalve, S., Cufino, F., Kose, U., Mascellani, A. & Rubbia, A. faserDL: Code for "Towards foundation-style models for energy-frontier heterogeneous neutrino detectors via self-supervised pre-training". *Zenodo* https://doi.org/10.5281/zenodo.22011299 (2026).

Acknowledgements

Neural network training was supported through the Swiss AI Initiative via a grant from the Swiss National Supercomputing Centre (CSCS), project ID a0178, on the Alps system.

Author contributions

S.A.-M. and F.C. conceived the study and developed, trained and tested the deep learning workflow. S.A.-M., F.C. and A.M. performed the data analysis. U.K. and A.R. developed and ran the simulations. All authors contributed to writing the manuscript.

Funding
The authors received no specific financial funding for this work. Computational resources for neural network training were provided through the Swiss AI Initiative via the Swiss National Supercomputing Centre (CSCS), project ID a0178, on the Alps system. Open access funding provided by Swiss Federal Institute of Technology Zurich.

Competing interests
The authors declare no competing interests.

Additional information
Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s42256-026-01309-6.

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Peer review information *Nature Machine Intelligence* thanks the anonymous reviewer(s) for their contribution to the peer review of this work. Peer reviewer reports are available.

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Last updated by author(s): Aug 19, 2026

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